TOPIC 4.3

How has technology expanded our knowledge and understanding of the universe?

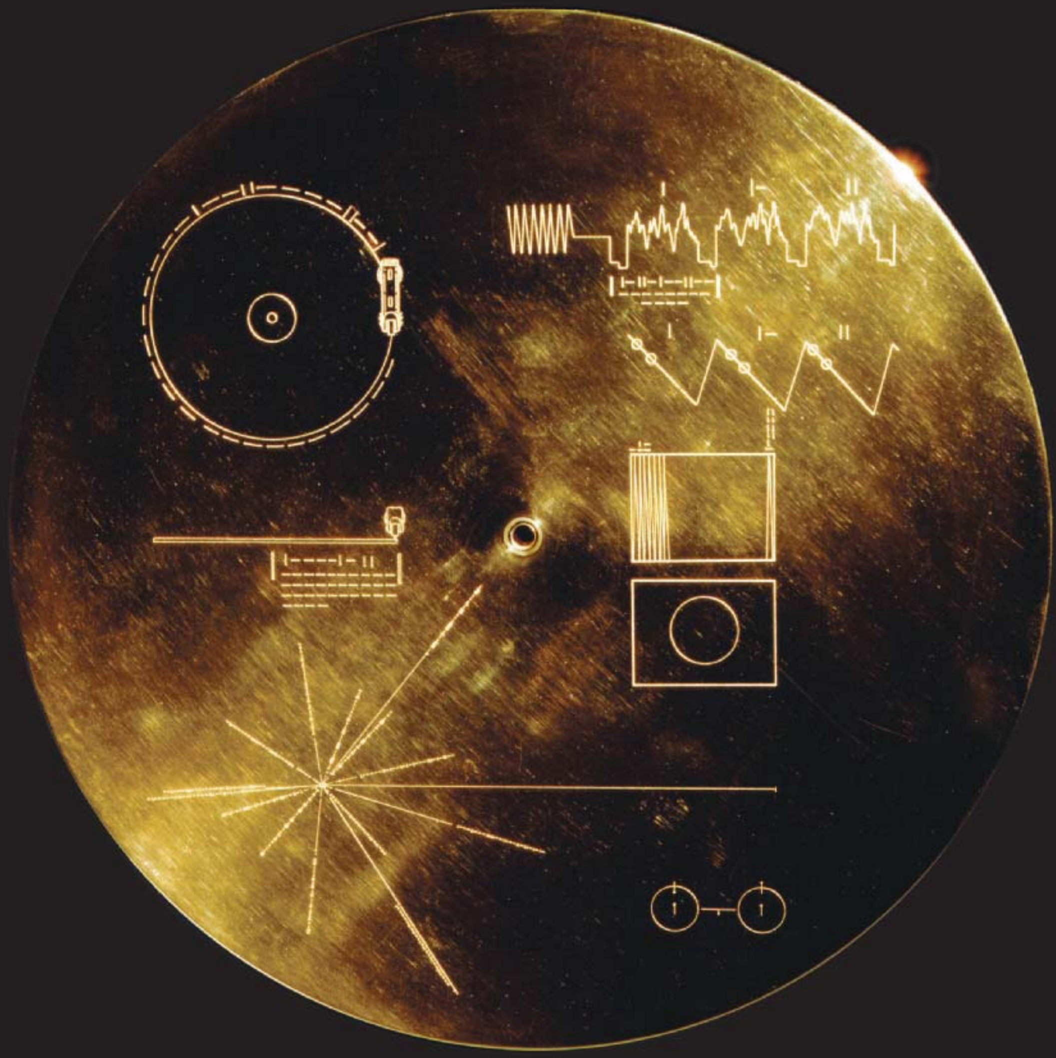
Key Concepts

- We have invented telescopes and other devices that extend and enhance our sense of sight.
- We know that our Milky Way galaxy is just one of many billions of galaxies in the universe.
- There are vast distances separating stars and separating galaxies.
- The properties of stars help us develop an understanding of their life cycles.

Curricular Competencies

- Construct, analyze, and interpret graphs, models, and diagrams.
- Analyze cause-and-effect relationships.
- Formulate physical or mental theoretical models.

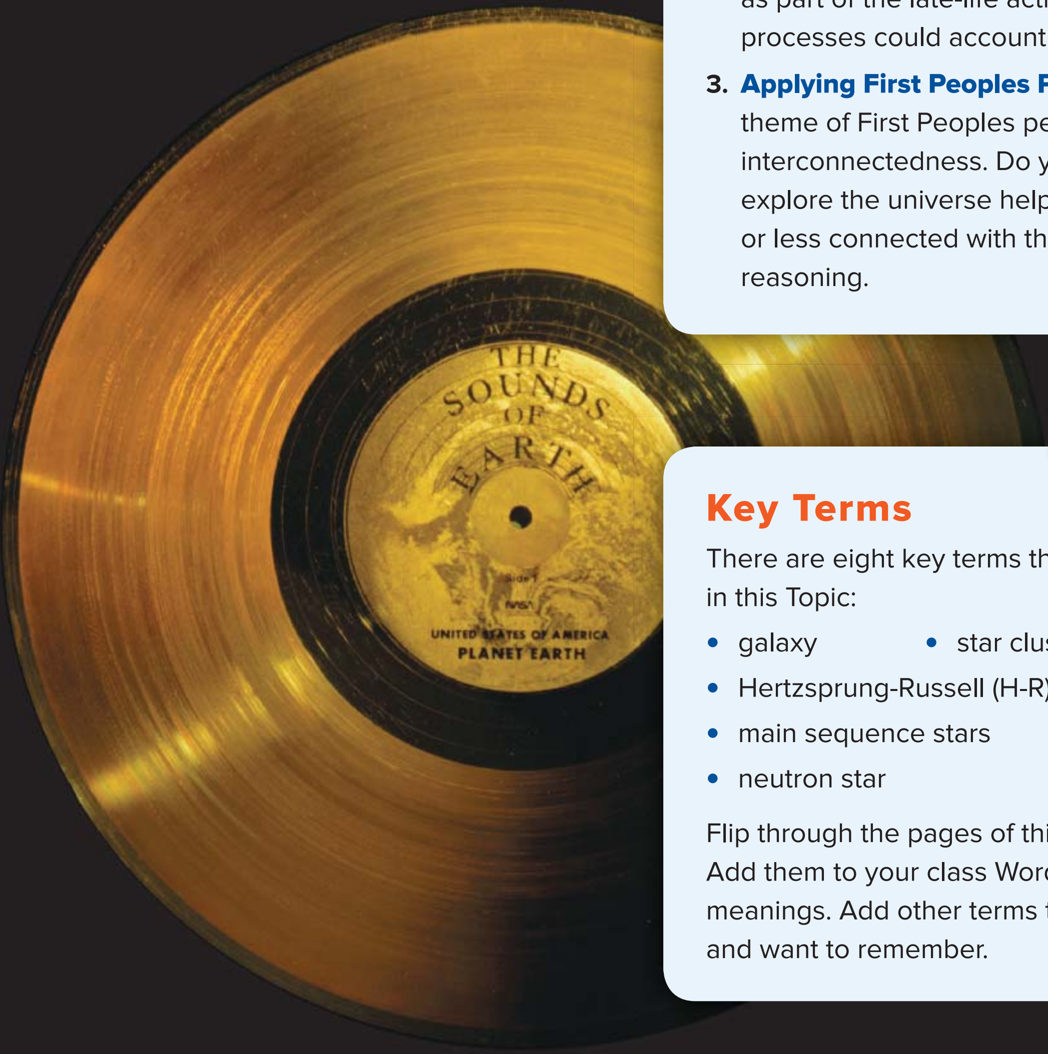
At this moment, more than 140 times the distance between the Sun and Earth, and counting, the *Voyager 2* spacecraft is hurtling at nearly 58 000 km/h through interstellar space. It carries gold-plated greetings, sights, and sounds from home, from us. And there is perhaps some irony in the fact that its lustrous gold came originally from the distant space into which it travels. Gold is thought to have been forged in cataclysmic cosmic events involving certain types of stars or star remnants in the late stages of their lives.



Starting Points

Choose one, some, or all of the following to start your exploration of this Topic.

1. **Communicating** You can access samples of the Voyager probe's golden record on the Internet. Because it was launched in 1977, many of the recordings on the golden record could fairly be called "golden oldies" by today's standards. If you were asked to create a new golden record to send into space, what sound, songs, and sights would you include? What would you be hoping to communicate to the universe with your choices?
2. **Questioning** How could gold (or any other element) form as part of the late-life activity of a star? What forces and processes could account for this?
3. **Applying First Peoples Perspectives** A key theme of First Peoples perspectives of science is interconnectedness. Do you think using technology to explore the universe helps us feel more connected or less connected with the universe? Explain your reasoning.



Key Terms

There are eight key terms that are highlighted in bold type in this Topic:

- galaxy
- star cluster
- light-year
- Hertzsprung-Russell (H-R) diagram
- main sequence stars
- supernova
- neutron star
- black hole

Flip through the pages of this Topic to find these terms. Add them to your class Word Wall along with their meanings. Add other terms that you think are important and want to remember.

CONCEPT 1

We have invented telescopes and other devices that extend and enhance our sense of sight.

Our exploration of the universe began with and depends on our sense of sight. As new technologies such as telescopes were invented, people began to use these tools to explore in new ways. By the time of Galileo Galilei in the early 1600s, the telescope had been invented. Galileo improved and turned this new instrument to the night sky and became the first person to observe moons around Jupiter and the rings of Saturn.

Telescopes of the type that Galileo used are called optical telescopes. They collect visible light from over a large area and concentrate it to form a brighter image. However, visible light is only one form of electromagnetic radiation that is given off by stars and other objects in space.

The ideal telescope would be able to detect not only visible light, but also radio waves and other forms of electromagnetic radiation. Such a device does not yet exist, but engineers have designed distinct telescopes that can detect non-visible radiation. One example is a radio telescope such as the one shown in **Figure 4.17**. Radio telescopes detect radio waves. Long-wavelength radio waves can penetrate clouds, so an advantage of using radio telescopes over optical devices is that they can be used on cloudy days as well as at night.

Figure 4.17 This 26 m radio telescope is part of the Dominion Astrophysical Observatory near Penticton. The observatory is owned and operated by Canada's National Research Council.

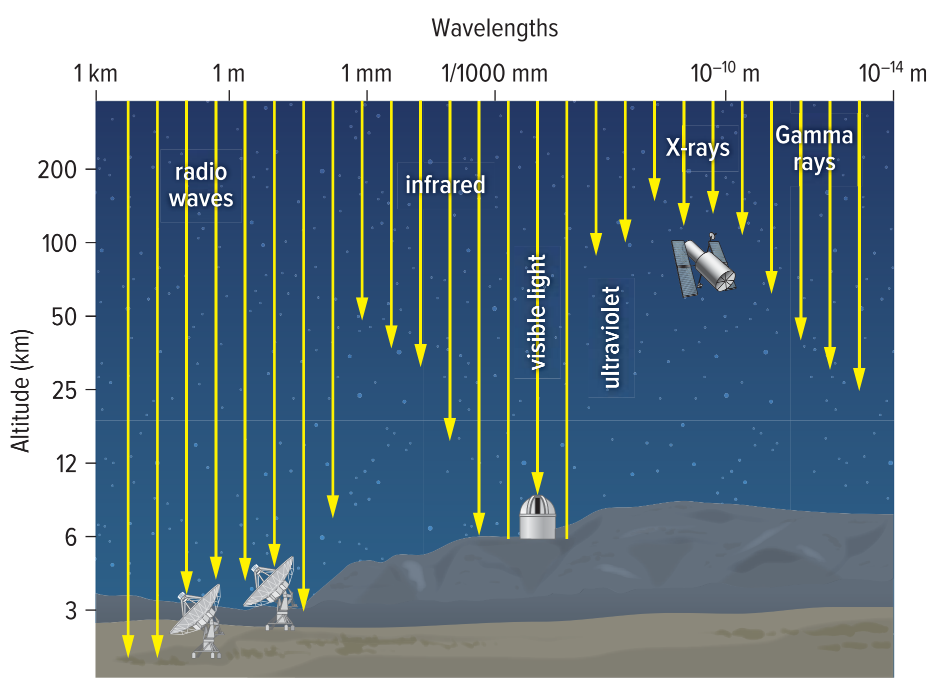


Figure 4.18 Only a fraction of electromagnetic radiation from space reaches Earth's surface. Therefore, some telescopes must be placed in orbit above the atmosphere to take advantage of the wealth of information the whole spectrum of this radiation provides.

Telescopes in Space

Much of the radiation that reaches Earth from space is absorbed by the atmosphere and does not reach the surface (**Figure 4.18**). For example, infrared radiation with a wavelength of 1 mm is absorbed about 50 km above Earth's surface. In order to observe space in more detail, some telescopes need to be placed above Earth's atmosphere. Some examples are the Chandra X-ray Observatory, the Spitzer Space Telescope (which detected infrared), and the Hubble Space Telescope. Hubble detects visible light as well as infrared and ultraviolet.

Studying Objects in Different Wavelengths

A range of telescopes can reveal different types of information about an object. For example, **Figure 4.19** shows the planet Saturn in four different wavelengths. Comparing photo **A** and **B** reveals that Saturn has auroras (the glowing regions at the poles in **B**), just as Earth does. The infrared view in **C** shows detailed features in Saturn's atmosphere. The different colours represent different heights and compositions of the planet's clouds. Photo **D** shows that the planet gives off radio waves (the red portion) and that the rings (the blue) absorb this radiation.

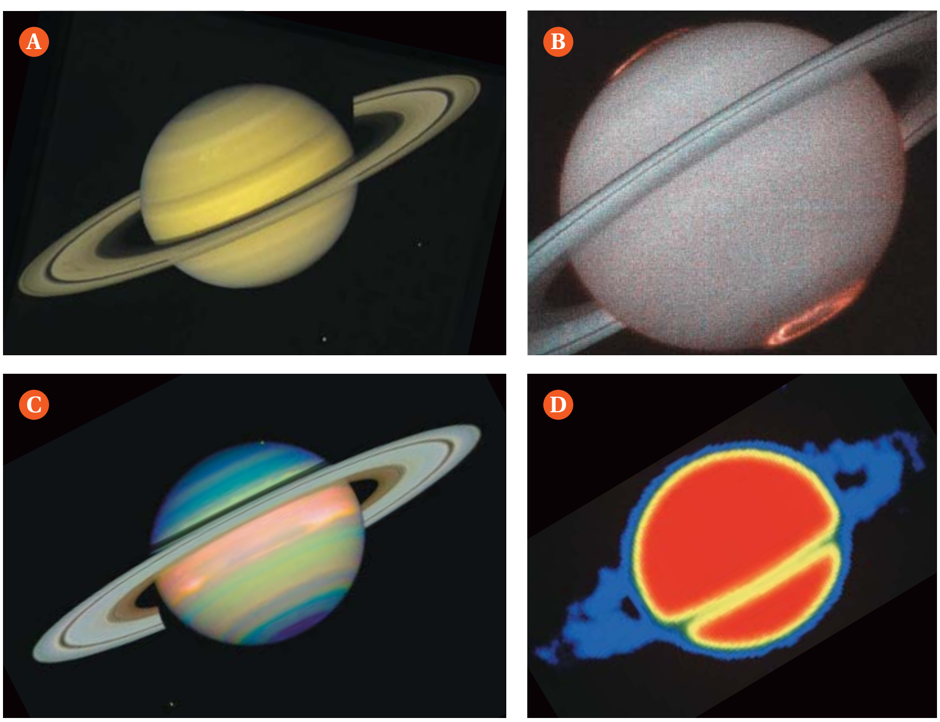


Figure 4.19 Saturn as viewed in visible light A, ultraviolet B, infrared C, and radio waves D. The colours in B, C, and D are false. They are added to highlight different features. **Applying: How does viewing the same object with different wavelengths affect our ability to understand it?**

Before you leave this page . . .

1. How did an optical telescope change the way that we can observe and explore space?
2. How do telescopes that detect non-optical radiation contribute to our understanding of space?

AT ISSUE Who has the MOST to lose?

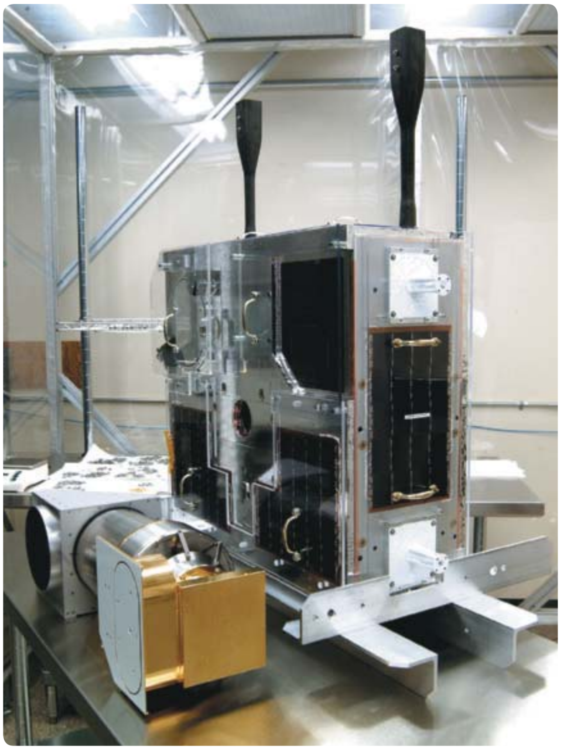
What's the Issue?

It may be tiny and economical as space telescopes go. But an innovative Canadian satellite is making big discoveries that are currently out of reach of even the powerful Hubble Space Telescope. It is also helping to change the future of satellite design. MOST (Microvariability and Oscillation of Stars) is Canada's first space telescope. Launched in 2003, its mission is to uncover the secrets behind the life cycles of Sun-like stars.

With its onboard telescope, MOST measures tiny changes in the brightness of stars. Although the telescope is the size of a dinner plate, it is so sensitive that it can detect a change in the brightness of a star as small as one part per million. That is like looking at a skyscraper lit up at night and measuring the change in overall brightness when someone lowers a shade at one window by 3 cm. No other space telescope can do that.

MOST is able to monitor the same star for seven weeks and then send the data to Canadian-based ground stations. In comparison, the Hubble telescope can keep a star in view for only six days.

MOST has discovered new data that give astronomers clues about the Sun in its youth. For 29 days, MOST observed a "teenage" Sun-like star in the constellation Cetus (Whale), as well as a middle-aged Sun-like star in Canis Minor (Little Dog). MOST's observations support predictions of what we think the Sun was like when it was young.



With a mass of only 56 kg, MOST belongs to a class of compact satellites called microsatellites that are being placed in Earth orbit. These lightweight satellites cost much less to design, build, and launch than their larger counterparts.

The MOST satellite itself is about the size of a large suitcase, and it cost less than $10 million to develop and manufacture. In comparison, its larger cousin, the Hubble Space Telescope, is about the size of a school bus and cost more than $2 billion to develop. Astronomers and satellite engineers from the Canadian Space Agency, the University of British Columbia, and the University of Toronto, as well as Mississauga, Ontario-based company Dynacon Enterprises, worked together to design and build MOST.

Mission Over—Or Is It?

The MOST mission was originally planned to last one year, but its successes led to an extended life. Even so, in 2014, funding support was ended by the Canadian Space Agency. This did not deter UBC astrophysicist and MOST team leader Jaymie Matthews, who searched for and eventually secured a MOST excellent buyer: Microsat Systems Canada Inc. Astronomers are able to pay a relatively modest (by astronomy standards) sum to use the micro-satellite for a week or more.

Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

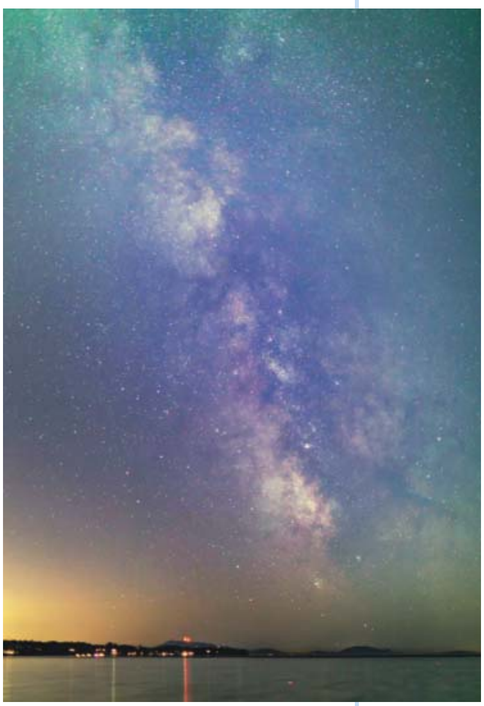
1. What are some of MOST's most impressive success stories?
2. Canada is a leader in developing miniaturized satellites. Its NEOSSat (Near Earth Object Surveillance Satellite) uses the same technology as MOST but is part of Earth's early warning system for asteroids and comets that could pose a threat to Earth in future. What other technologies and/or missions feature Canadian ingenuity and solutions?
3. Private aerospace companies such as SpaceX and Blue Origin both supplement and compete with publically funded organizations such as NASA and the European Space Agency. How has this commercialization affected the development of space-related science and technology?
4. What other space telescopes are currently in use or planned for the future? Which portion of the electromagnetic spectrum will they use to view objects, and what are the goals of the missions?

CONCEPT 2

We know that our Milky Way galaxy is just one of many billions of galaxies in the universe.

If you have viewed the night sky in a dark place, far from intruding urban lights, you have likely seen the Milky Way. Brightest in the summer, it appears as a hazy white band extending from the southern horizon and across the sky overhead (Figure 4.20). In fact, the band is a vast accumulation of about 400 billion stars that completely encircles Earth. The Milky Way is a **galaxy**—a collection of stars, gas, and dust held together by gravity. (The gas of a galaxy is made up mainly of hydrogen atoms. The dust is not like dust on Earth. Instead, space dust is made up of carbon and silicate particles about 100 nm in size.)

galaxy a collection of many billions of stars, plus gas and dust, held together by gravity



The Discovery of Galaxies

In 1610, using his telescope, Galileo became the first person to realize that the fuzzy Milky Way band was actually a collection of individual stars. It took another 170 years before further understanding of these stars developed. At that time, a British astronomer, William Herschel, was exploring the heavens. He and his sister, Caroline, were famous for building and selling fine telescopes. Together, using their inventions, they discovered that the Milky Way is a gigantic system of stars—what we now call a galaxy. Every star that you see in the sky on a clear night is part of the Milky Way galaxy.

Figure 4.20 In a dark sky, on a clear night, the Milky Way galaxy looks like a band of white in the night sky. The ancient Romans called it the *Via Lactea*, which means "way (or road) of milk." The term "galaxy" also comes from an ancient word (Greek, this time), *galactos*, meaning milk.

Activity

Model Galaxy Motion

Add about 250 mL of water to a 500 mL beaker. Hold it carefully, and swirl it slowly to give the water a circular motion. Put the beaker down, with water still swirling, and add a few drops of food colouring to the centre. Sketch your observations.

Repeat this process with water swirling faster and slower, and with a pinch of light, dry material such as dried oregano or cumin. Compare how the different materials behave in the swirling water. How is this model similar to real galaxy motion? What are the limitations of this model?

The Shapes of Galaxies

A galaxy forms when gravity causes a large, slowly spinning cloud of gas, dust, and stars to contract (draw together). The Sun is one of an estimated 400 billion stars in the Milky Way. All the stars in the universe belong to one of the billions of galaxies that exist.

Galaxies come in different shapes and sizes. Generally, they are classified as either elliptical or spiral, according to their appearance. Galaxies that do not fit into these general classifications are called irregular galaxies.

Elliptical galaxies vary in shape from spherical to a flattened oval (**Figure 4.21A**). They are older galaxies with very few young stars. Ellipticals account for 15% to 20% of all galaxies we can see.

Spiral galaxies look like pinwheels—flattened disks with a central bulge and two to four spiral arms (**Figure 4.21B**). Their central core is made of up old stars. The spiral arms contain clouds of gas and dust along with new and young stars. Our Milky Way is a spiral galaxy. A subclass of spirals are barred spiral galaxies (**Figure 4.21C**). They are similar to spiral galaxies, but they have a central bar pattern across the middle, with spiral arms trailing from the ends of these bars.

Irregular galaxies have no definite shape (**Figure 4.21D**). They contain more gas and dust than their spiral cousins. They have no spiral arms or central core, and they make up at least 10% of all galaxies.

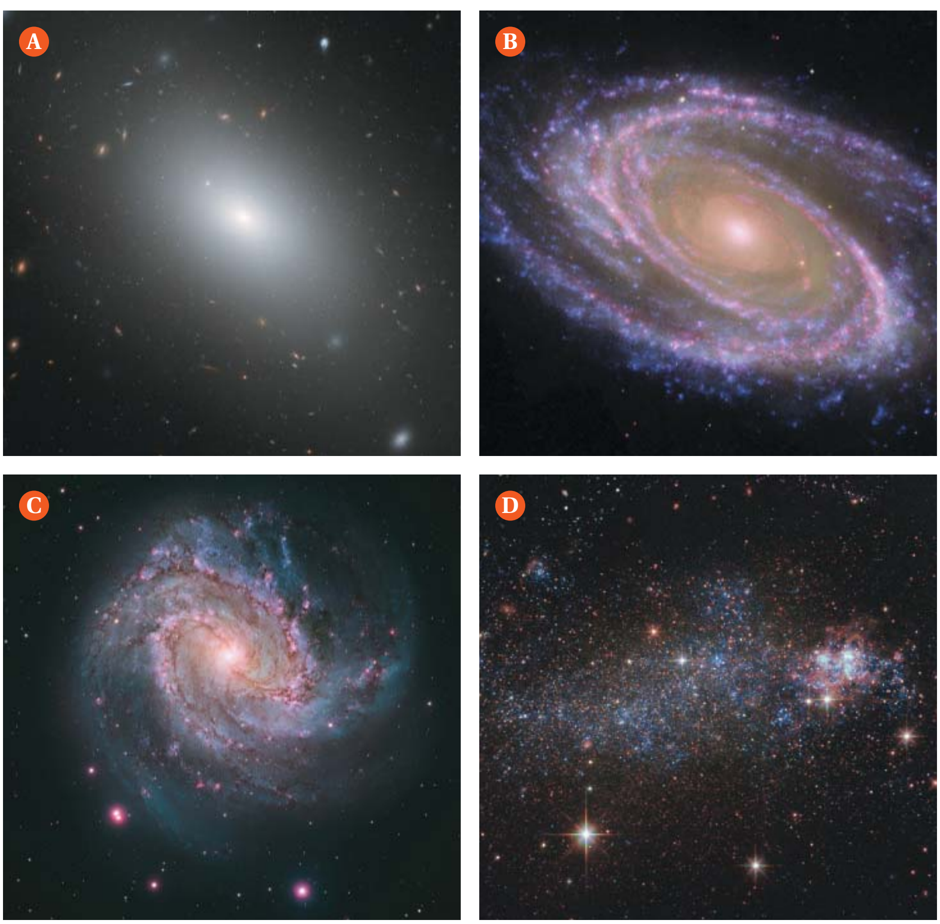


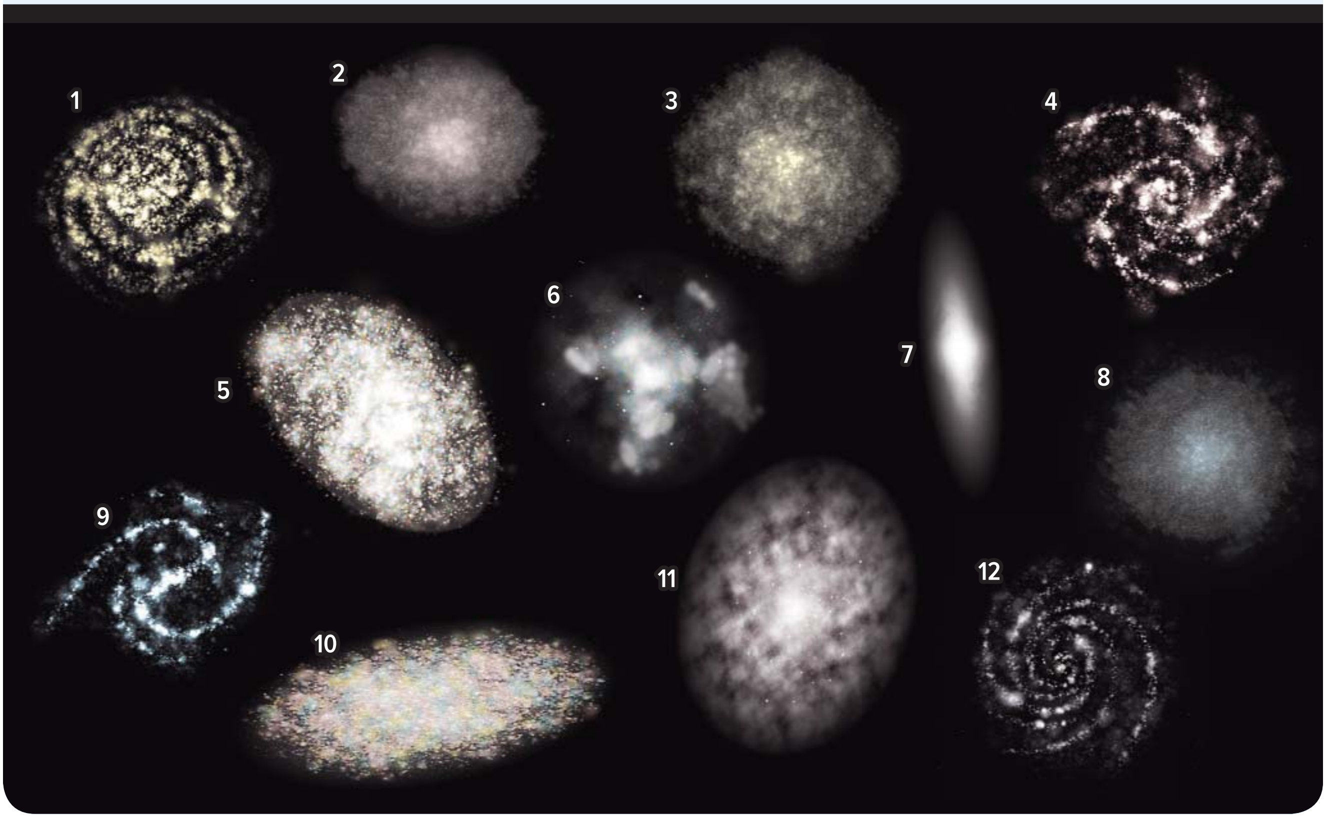
Figure 4.21 Main types of galaxies—an elliptical A, spiral B, barred spiral C, and irregular D.

Activity

Picturing Galaxies

Although astronomers can classify galaxies into different types based on their shape, it can be difficult to get clear images of their shape. Galaxies can be blocked by gas and dust or other celestial objects. Even when we get a clear view, the galaxy can be difficult to classify because of the angle we are observing it from.

1. Your teacher will give you four blank plastic sheets. Draw one type of galaxy in as much detail as you can on each of the sheets.
2. Have a classmate hold the sheets up one at a time on the other side of the room so they are facing you. Identify each type.
3. Have your classmate tilt each of the sheets at various angles while you try to identify the galaxy types.
4. Finally, have your classmate hold each sheet parallel to the floor while you try to identify the galaxy types.
    - Were the galaxies easy to identify on the other side of the room when they were facing you?
    - As your classmate tilted the sheets, did it become easier or harder to identify the types of galaxies?
    - When the sheets were parallel with the floor, could you see any detail that might help you decide what kinds of galaxies they were?
    - Explain how the angle of observation might make it difficult for astronomers to distinguish spiral galaxies from barred spiral galaxies.
    - The image below represents a random sample of galaxies from the universe. Decide how you would classify each galaxy, and give or discuss reasons in each case.



Understanding the Milky Way

It has taken astronomers many years to learn about the Milky Way galaxy. William Herschel started putting together the pieces of the puzzle. By counting stars, Herschel figured out the approximate shape of the Milky Way galaxy. Herschel proposed that the Milky Way is a huge disk of billions of stars, flattened like a dinner plate, in which the Sun is embedded. He also proposed that the Sun might be at the centre of the Milky Way.

Star Clusters

In the early 20th century, American astronomer Harlow Shapley helped to put together more pieces of the puzzle. While studying the Milky Way, he was also studying **star clusters**. A star cluster is a collection of stars held together by its own gravity. Like individual stars, star clusters are held inside or around galaxies by gravity. **Figure 4.22** shows the two types: open clusters and globular clusters. Open clusters have 50 to 1000 stars and appear along the disk of the Milky Way. Globular clusters have a spherical shape with 100 000 to 1 000 000 stars.

Shapley became interested in globular clusters in particular. He reasoned that globular clusters should be evenly distributed around the galaxy. He noticed, however, that they appear only in the direction of the constellations Hercules, Scorpius, Ophiuchus, and Sagittarius—not all around us. Shapley reasoned that his observations could only be explained if he were observing the globular clusters from a position well away from them. He concluded that if globular clusters were spread out around the Milky Way's centre, the Sun must be nowhere near the centre.

star cluster a collection of stars held together by gravity

The Diameter of the Milky Way

Radio waves can travel through the clouds of Earth's atmosphere. They also can travel through the gas and dust between stars. By mapping the galaxy with radio waves, astronomers have been able to determine that its shape is disk-like and that its diameter is about 100 000 light-years across. (A light-year is the distance that light travels in one year: 9.5×10^{12} km. You will learn more about light-years and distances within and between galaxies in the next Concept.)

Figure 4.22 The Pleiades open cluster A is found in the constellation Taurus. Globular clusters B contain many more stars than open clusters do.

The Centre of the Milky Way Galaxy

Using radio waves as well as infrared radiation, astronomers next confirmed that the centre of the Milky Way galaxy is surrounded by a bulge of stars. Around the bulge, there is a sphere of globular clusters, as shown in **Figure 4.23**. When Shapley was observing globular clusters, he was looking toward the centre of the Milky Way galaxy and the halo of globular clusters that surround the galaxy from a position in the disk well away from the centre.

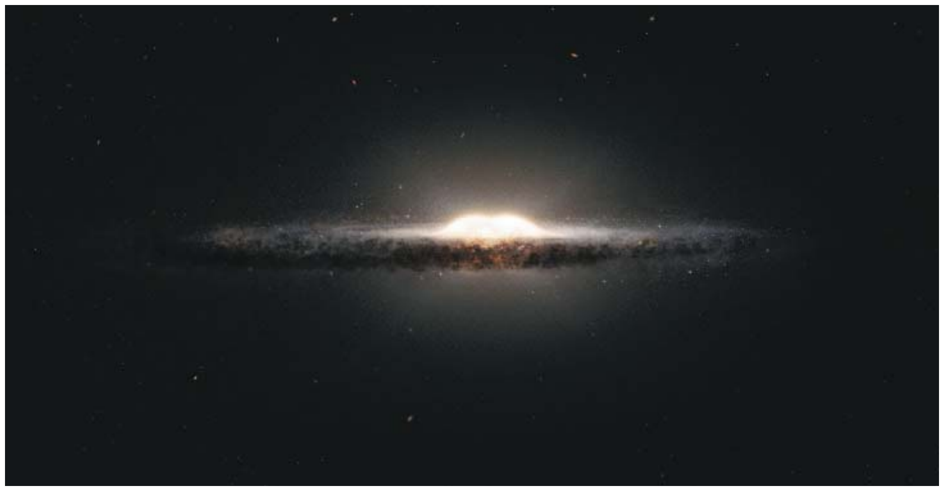


Figure 4.23 Globular clusters form a sphere around the centre of the Milky Way.

Some other Characteristics of the Milky Way Galaxy

Knowing that the Milky Way galaxy has a disk-like shape, with a central bulge of stars, astronomers have concluded that it is a spiral galaxy.

Using data from the European Space Agency's (ESA's) Gaia space telescope, which is creating a 3D map of our galaxy, astronomers from the University of Toronto have calculated our Sun's distance from the galactic centre to be 24 788 to 26 745 light years. With telescopes that use different parts of the electromagnetic spectrum, astronomers can also image various regions of the Milky Way (**Figure 4.24**).

Figure 4.24 The dark, reddish-brown areas across the centre of this image of the Milky Way galaxy are called lanes. This darkened zone is caused by enormous clouds of gas and dust that are blocking the light from the background stars in the galaxy. Using other electromagnetic frequencies enables astronomers to "see through" the lanes.

Galaxy Groups and Clusters

The Milky Way belongs to a group of about 50 galaxies called the Local Group. Some are shown in **Figure 4.25**. The diameter of the Local Group is about 10 million light-years. Our nearest comparable-size galactic neighbour is the spiral-shaped Andromeda galaxy. It lies about 2.6 million light-years away and is the farthest object in the sky that we can see with the unaided eye. The earliest recorded observation of the Andromeda galaxy was made by Persian-Islamic astronomer Abd al-Rahman al-Sufi. He described it as a fuzzy cloud in his famous *Book of Fixed Stars* in the year 964. Our galaxy is estimated to collide with the Andromeda galaxy in about 4 billion years.

Our Local Group belongs to a much larger collection of galaxies called the Virgo Supercluster. And this is just one of millions of superclusters in the universe!

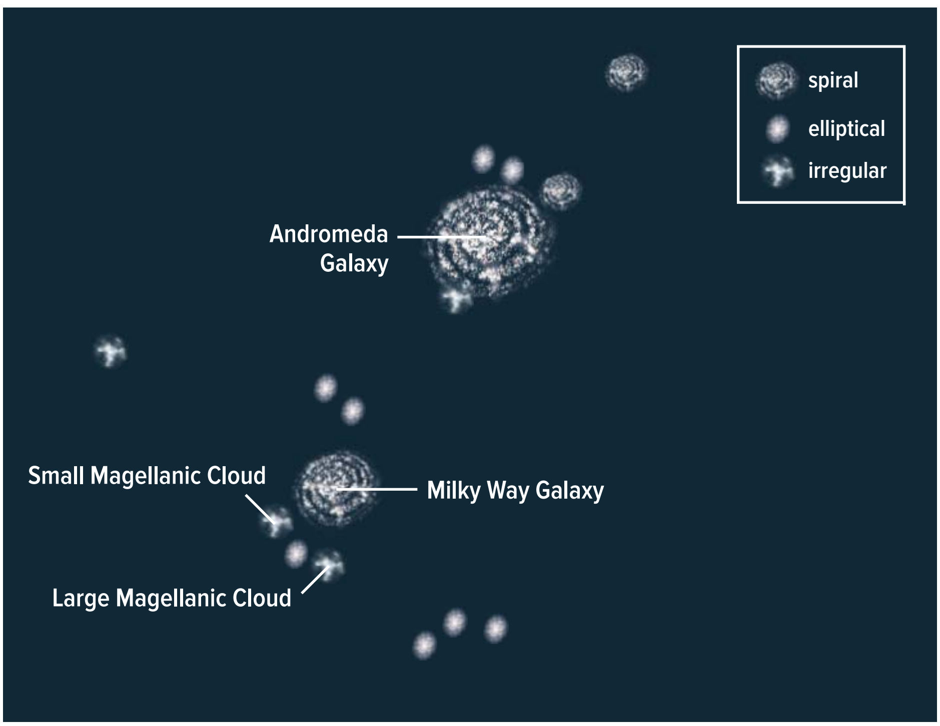


Figure 4.25 The Local Group, made up of many small irregular and elliptical galaxies, is dominated by two large spiral galaxies: our Milky Way and Andromeda.

Before you leave this page . . .

1. Compare the three basic shapes and sizes of galaxies.
2. How did we infer our position within the Milky Way galaxy?
3. How did we infer that the Milky Way is a spiral galaxy?
4. What is the Local Group?

Extending the Connection

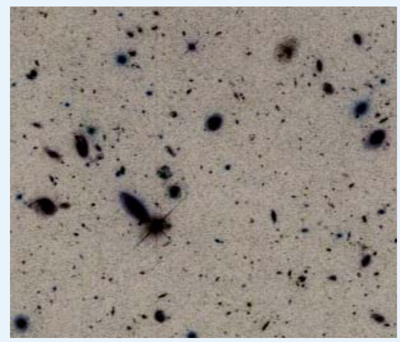
Determining our Distance from the Galactic Centre

FInd out how U of T astronomers used Gaia to measure the speed and distance of our Sun's orbit around the centre of the Milky Way.

Activity

Counting Galaxies by Sampling

Have you ever wondered how scientists know how many hairs are on the average cat? Counting every hair on a cat would take a very long time. A mature domestic cat has about 3 million hairs. It would take more than a month, counting one hair per second, 24 h a day, non-stop, to count every hair. A more feasible (and realistic) strategy is to estimate the number using a technique called sampling. In this activity, you will use sampling to estimate the number of galaxies that the Hubble Space Telescope (HST) can observe in a small area of the sky.



Galaxies as viewed with the Hubble Space Telescope. The image inverts black and white to make it easier to count the galaxies.

1. Your teacher will give you a copy of the image shown here. Each group will study one section of the image.
2. Tally the number of galaxies in your section. Every small smudge on the image is a galaxy, except the smudges that have "spikes" radiating from them. The spiked smudges are stars. (The spikes are interference effects caused as light travels through the telescope.)
3. Collect the tallies for the other sections, add all the tallies together, and calculate the average number of galaxies per section.
4. Astronomers use degrees to represent measurements in the sky. For example, when measuring in degrees, the diameter of the full Moon is one half a degree, where the angle between the east horizon and the west horizon is 180°. One section in the image represents approximately 2.2×10^{-4} square degrees in the sky. The total area of the sky is 4.13×10^4 square degrees. Estimate the total number of galaxies in the universe as follows:

$$\text{Total number of galaxies} = \frac{\text{total area of the sky in square degrees}}{\text{area of one section} \times \text{average number per section}}$$

$$= \frac{4.13 \times 10^4 \text{ square degrees}}{2.2 \times 10^{-4} \text{ square degrees} \times \text{average number per section}}$$

- Approximately how many galaxies can HST see in this image?
- Why is it more accurate to average the tallies for all the sections to get the average number per section rather than use the results for a single section?
- Why is sampling more practical than trying to do a more detailed count?

Focus on Astronomy

What kinds of jobs involve astronomy?

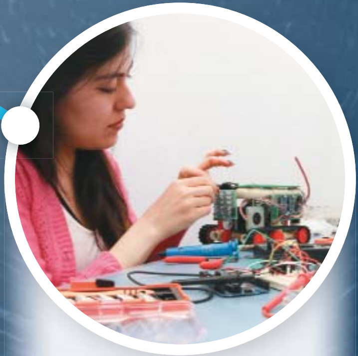
- Astronaut
- Radar Technician
- Satellite Communications Operator
- Aerospace Engineer
- Astrophysicist

Space Educator

Space educators work at schools, museums, and planetariums like the one at Vancouver's H.R. MacMillan Space Centre. They need a solid understanding of astronomy and a talent for teaching or presenting.

Atmospheric Scientist

Some atmospheric scientists monitor disturbances in Earth's atmosphere due to space weather, such as solar radiation storms. They issue space weather alerts when storms threaten Earth's power grids, satellites, and radio communications.



Robotics Engineer

Robotics engineers design and build automated machines like the Canadarm2. This robotic arm is attached to the International Space Station. It helps transport supplies and equipment in space.

Questions

1. What other jobs and careers do you know or can you think of that involve astronomy?
2. Research a job or career related to Unit 4 that interests you. What attracts you to it? What kinds of things do you have to know, do, and understand for this job or career?

CONCEPT 3

There are vast distances separating stars and separating galaxies.

For distances on Earth, the kilometre serves us well. Within our solar system, it is more convenient to use the astronomical unit (AU). One AU is the distance between the Sun and Earth: 150 000 000 km. Thus, the Earth-Sun distance is 1 AU (**Figure 4.26**.) The distance from the Sun to the last planet of the solar system, Neptune, is about 30 AU. However, distances beyond the solar system are too great for AUs, since the distance to the closest star is more than 268 000 AU. This would be like measuring the distance from Victoria to Hope in metres.

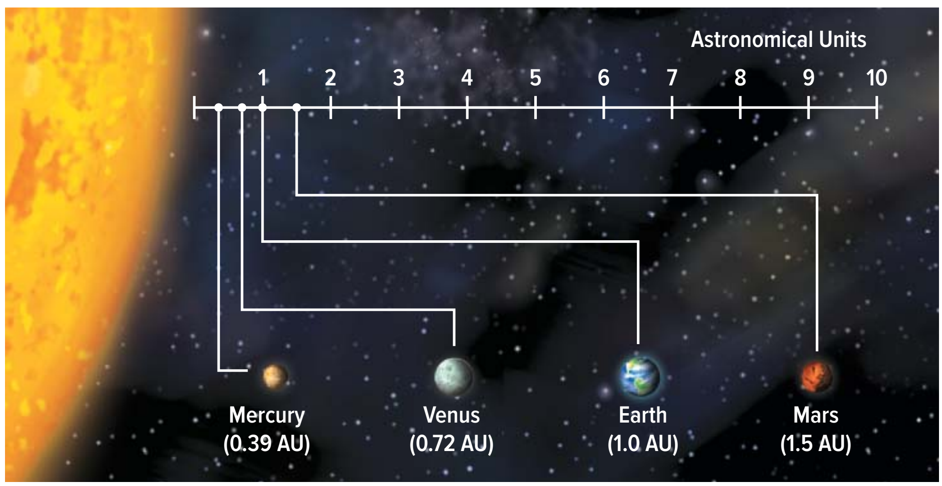


Figure 4.26 Using the AU to measure distances in the solar system is simpler and more convenient than using kilometres.

The Light-Year

Stellar and interstellar distances are typically measured in **light-years**. A light-year is the distance that light travels in a vacuum (empty space) in one year. In space, light travels at a constant speed of about 300 000 km/s. So 1 light-year (ly) is approximately equal to 10 trillion (9.46×10^{12}) km. **Table 4.1** shows the distances of several stars and galaxies.

light-years a unit of distance equal to the distance light travels in one year

Table 4.1 Distance of Some Celestial Objects

Star or Galaxy	Approximate Distance from Earth (ly)
Proxima Centauri (star closest to ours)	4.24
Vega (part of the constellation Lyra)	25
Polaris (part of the constellation Ursa Minor)	400
Betelgeuse (part of the constellation Orion)	643 (plus or minus 46)
Deneb (part of the constellation Cygnus)	1400
Andromeda Galaxy (galaxy nearest to ours)	2 600 000

Extending the Connections

Why don't we know the distance to Betelgeuse?

Did you notice that value for Betelgeuse in Table 4.1? What are your questions, and how can you answer them?

Implications of the Light-Year

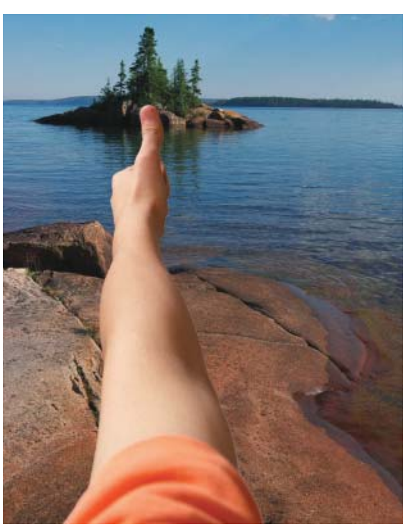
The farther a star is from Earth, the longer it takes for its light to reach us. For example, Vega is about 25 ly from Earth. The light from Vega travels at the speed limit of the universe—the speed of light. This means that it takes the light from Vega about 25 years to reach us here on Earth. When you observe Vega in the night sky, you see the light that has taken 25 years to reach your eyes. In other words, you see Vega as it was 25 years ago, not as it is today. It is, in fact, impossible to see Vega in "real time." You can't even see the Sun in real time. The light that travels from the Sun to Earth takes 8 minutes to reach us. The Sun is always 8 minutes old to our eyes. Whenever we look at celestial objects, with or without technology, we see what was, not what is. We are looking back in time.

Stellar Distances

Early astronomers discovered that over a period of weeks and months, the planets appeared to move against a background of stars. This led them to believe that stars are much farther away than the planets. However, we did not have a useful way to measure the distance between celestial objects or the distance between these objects and Earth.

Scientists have since developed various methods to measure interstellar distances—the distances between stars in our galaxy. One method, called *parallax*, relies on geometry. Parallax is the apparent change in position of an object against a fixed background when it is viewed from two different lines of sight. You can observe parallax using the method described in Figure 4.27.

A



B

Figure 4.27 In A, a student looks through only his right eye as he lines up his thumb with an island in the distance. In B, the student views the same island through only the left eye, and the island seems to change its position. **Conducting: Try this out for yourself!**

Explaining Parallax

Figure 4.28 presents a different view of the phenomenon described in **Figure 4.27**. In **Figure 4.28**, each eye views your thumb (the object) from a slightly different position. This results in an apparent change in the position of your thumb when you close one eye. When you have both eyes open, your brain uses parallax to get a sense of the distance between your eyes and your thumb. If you move your thumb halfway to your nose, you can see that parallax increases as distance decreases.

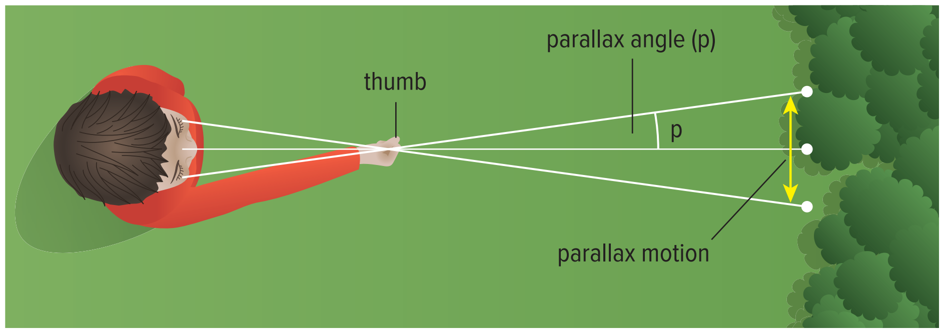


Figure 4.28 Parallax angle is the angle measured between two lines of sight, divided by two.

Parallax can be used to calculate the distance to stars using a method known as *triangulation*. With this method, if you know the length of one side of a triangle and the angles of two of the corners (vertices), you can calculate all the dimensions of the triangle.

Figure 4.29 shows how astronomers use triangulation to calculate the distance from Earth to a star. The diameter of Earth's orbit is the "known" side of the triangle. The star is then observed from two different positions in Earth's orbit around the Sun. To do this, astronomers observe the star and then wait six months to observe it again when Earth is at the opposite end of its orbit. The "triangle" formed by these observations can be used to calculate the distance to the star.

With the aid of extremely large telescopes, astronomers are able to use parallax to estimate the distances to stars as far as 100 ly away. For distances beyond this, using parallax becomes too imprecise. In the early 1990s, a unique satellite called HIPPARCOS (High Precision PARallax COllecting Satellite) was placed in orbit on a star-mapping mission. It was able to accurately measure star positions, parallaxes, and motions of stars up to 500 ly away. The ESA's Gaia telescope is measuring the parallax to about 1 billion stars in the Milky Way and other nearby galaxies.

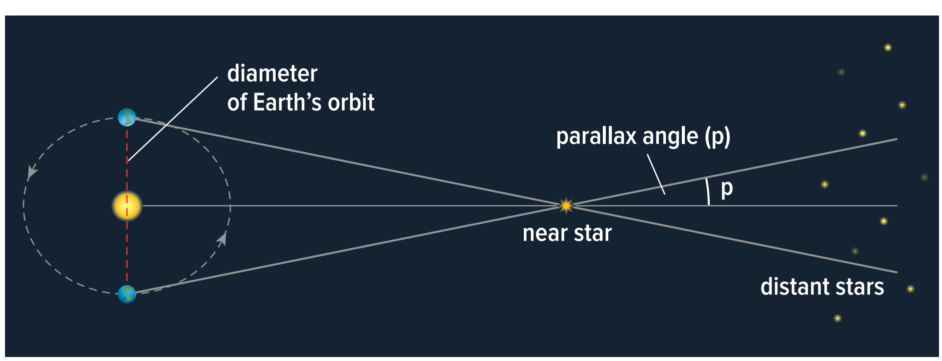


Figure 4.29 Because stars are distant, parallax must be determined from positions that are very far apart. **Innovating: Why is it ingenious to use Earth's orbit to determine the parallax of distant stars?**

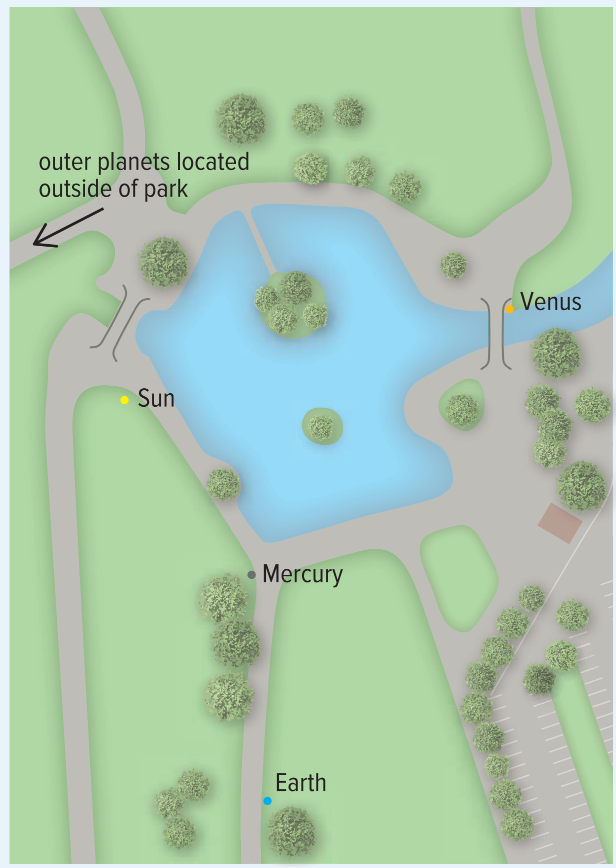
Activity

Cycle the Cosmos—A Local Community Distance Model

It began more than 25 years ago with a father's desire to help his fourth-grade child better understand the scale and structure of our neighbouring planets. Now, Alton Baker Park in Eugene, Oregon, is the site of a community-based model that enables people to walk (and ride) the distances separating the planets of the solar system. Similar models that map planets and planetary distances against places of interest have been designed in other communities around the world. What about yours?

In this activity, you are challenged to use your community as the foundation for a scale model of the solar system and beyond, far beyond. Go the distance—once you decide what the distance will be.

- What criteria will you consider to design a model that includes some of our interstellar neighbours?
- Is this a model that you can actually build in your community, or will you rely on virtual space to showcase it?
- Can you incorporate fitness, education, and/or other opportunities for community participation?
- What other decisions are needed to turn your ideas into reality?



Community-level scale model of the solar system at Alton Baker Park in Oregon: This model uses a scale of 1:1 billion and was designed to represent distances (and planet sizes) of the solar system. How can you include stars and galaxies that are more distant than the objects of our solar system?

Before you leave this page . . .

1. Describe a case where it might be convenient for astronomers to use **a)** AUs and **b)** light years.
2. Explain why kilometres are not adequate for measuring most distances in the universe.
3. The light-year is a unit of distance. Why does it seem like a unit of time?
4. How does parallax help us determine distance?

CONCEPT 4

The properties of stars help us develop an understanding of their life cycles.

Activity

Inferring Factors That Affect Brightness

Safety

Never use a laser pointer in place of a penlight or other small flashlight in this activity. Someone could be blinded.

For this activity, your group will use three flashlights—a small one (such as a penlight) and two larger ones identical in size and type.

1. One student stands at the back of the room holding one small and one large flashlight in each hand. At the front, with the room dark, have the student at the back turn on both flashlights. Record how their brightness compares.
2. Two group members stand side by side at the back, each holding one of the large flashlights. Again, stand at the front. Darken the room and have the students turn on the flashlights. Record your observations.
3. Have one student at the back walk toward you; the other stays still. Compare the brightness of the two lights. Have the approaching students use the small flashlight. How does the brightness of the two lights compare?
4. **a)** What factors affect the brightness of light? What might this suggest about the nature of stars and how bright they appear to be?
   b) Imagine two stars emit the same amount of light, but one appears to be 50 times brighter than the other. What might you infer about how far each star is from Earth?

Astronomers estimate that there are about 2 trillion galaxies in the observable universe. That's about 2×10^{23} stars. Like people, each star has unique properties such as brightness, colour, temperature, composition, and mass. These properties help astronomers understand the life cycles of stars—how they form and what can happen to them over their long and, in some cases, turbulent lives.

Brightness of Stars

Astronomers refer to luminosity when describing the brightness of stars. A star's luminosity is a measure of the total amount of energy it gives off per second. The star we know best is the Sun, so it is helpful to use it to compare with other stars. Astronomers have determined that some stars are about 10 000 times less luminous than the Sun, while others are more than 30 000 times more luminous.

Extending the Connections

Comparing Magnitudes

There are two scales that astronomers and other skywatchers use to describe star brightness. One, called apparent magnitude, was developed about 2200 years ago. The other, called absolute magnitude, is a much more recent invention. How do these scales work, and how do they compare? Why are both still in use?

Another term referring to a star's luminosity is absolute magnitude—how bright a star would be if it were at a distance of 32.6 light-years from Earth. The absolute magnitude of the Sun is about 4.7. This value reveals our Sun to be an average star in terms of luminosity. Some stars would be a million times brighter if they were as close to Earth as the Sun is. Others would be much more faint.

Colour and Temperature of Stars

The colour of a star gives scientists an indication of its surface temperature. A fairly hot star appears bluish in colour, while a fairly cool star appears reddish. See **Figure 4.30** and **Table 4.2**. Rigel appears bluish-white, while Betelgeuse appears reddish-white. Our Sun, with a surface temperature of 6000°C, appears yellowish, which places it midway between bluish and reddish stars.

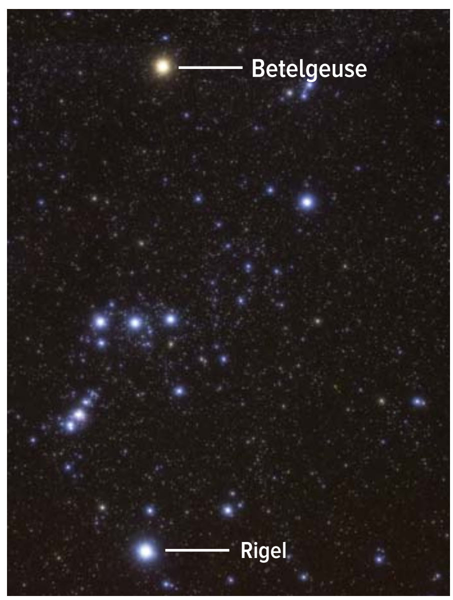


Figure 4.30 Comparing the colours and temperatures of two stars in the constellation Orion

Table 4.2 Colour and Temperature Ranges of Some Stars

Colour	Temperature Range (°C)	Examples
bluish	25 000–50 000	Zeta Orionis
bluish-white	11 000–25000	Rigel, Spica
whitish	7500–11 000	Vega, Sirius
yellowish-white	6000–7500	Polaris, Procyon
yellowish	5000–6000	Sun, Alpha Centauri
orangish	3500–5000	Arcturus, Aldebaran
reddish	2000–3500	Betelgeuse, Antares

Composition of Stars

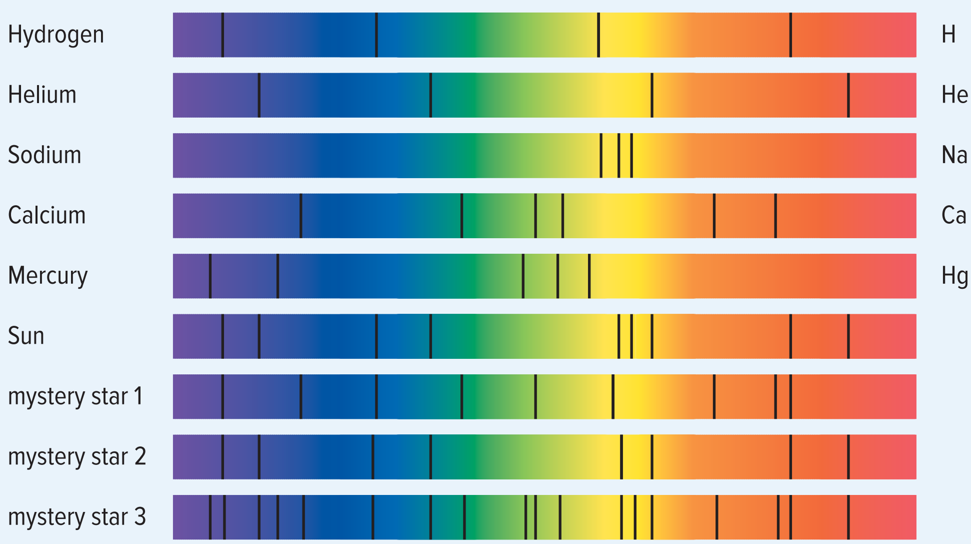
How do astronomers know that the Sun and other stars are made of hydrogen and helium? They use a spectroscope to analyze the light from stars. A spectroscope is an instrument that produces a pattern of colours and lines, called a spectrum, from a narrow beam of light. In the 1820s, Joseph von Fraunhofer, a German optician, used a spectroscope to observe the Sun's spectrum and noticed hundreds of lines. These lines are called spectral lines. He mapped the Sun's spectrum completely, although he did not know what the spectral lines meant. Today, we know that a star's spectrum identifies the elements of which it is composed (**Figure 4.31**).

Figure 4.31 Each element is uniquely identified by its spectrum.

Activity

Identify the Composition of Three Mystery Stars

Examine the spectral lines below. Compare the patterns of the known elements to those of the Sun and the three unknown stars.



- Which elements are present in the Sun's spectrum?
- In which two mystery stars is calcium present?
- Which mystery star contains sodium?
- Only one mystery star contains mercury. Which one is it?
- Which mystery star's composition is least like that of the Sun?
- Describe, in writing or orally with a partner, how you can infer a star's composition by analyzing the pattern of its spectra.

Mass of Stars

Determining the mass of stars was not possible until astronomers discovered that most of the stars seen from Earth are binary stars. Binary stars are two stars that orbit each other. The Sun is unusual in that it is not part of a binary star system. By knowing the size of the orbit of a binary pair and the time the two stars take to complete one orbit, astronomers can calculate the mass of each star. Star mass is expressed in terms of solar mass. The mass of the Sun is 10×10^{30} kg, which is given a value of 1 solar mass. Other stars range from 0.08 solar masses to over 100 solar masses.

The Hertzsprung-Russell Diagram and Stellar Evolution

As astronomers learned more about the properties of stars, they began to look for patterns in the data. In the 1920s, two astronomers were studying data from large numbers of stars visible from Earth. Ejnar Hertzsprung in the Netherlands and Henry Norris Russell in the United States were working independently of each other. But both observed that each star type has certain properties. These relationships can be shown on a graph called the **Hertzsprung-Russell (H-R) diagram**. Their graph had star colour (ranging from blue to red) on the *x*-axis. Absolute magnitude was on the *y*-axis of their graph. Astronomers discovered from the H-R diagram that there are several different categories of stars (**Figure 4.32**).

Connect to Investigation 4-C on page 348

Hertzsprung-Russell (H-R) diagram a graph that compares the properties of stars

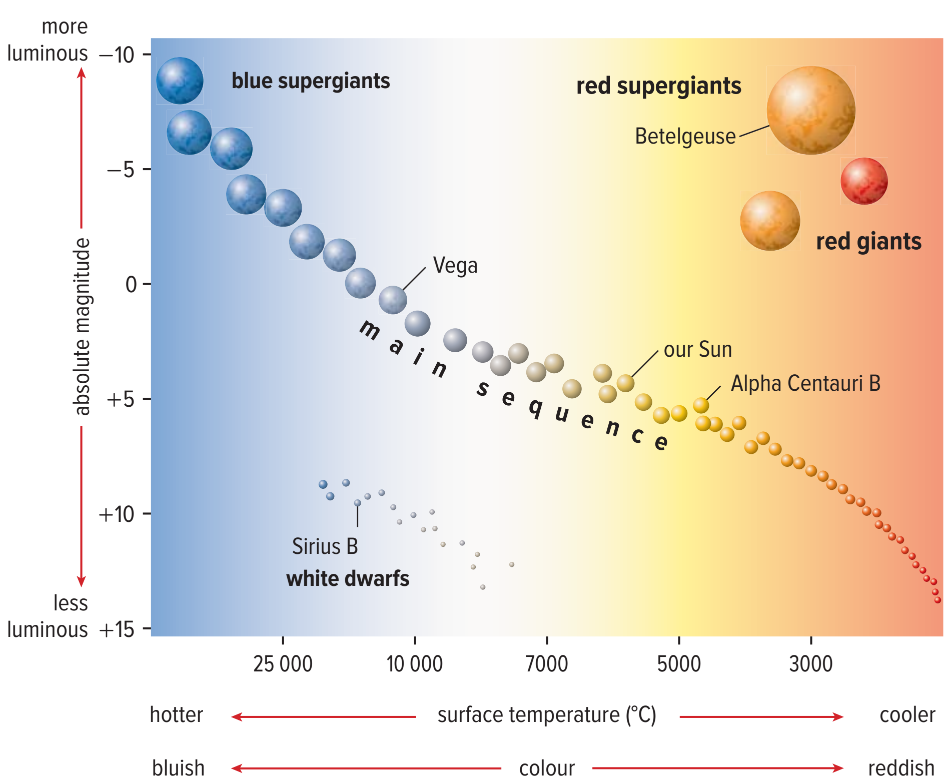


Figure 4.32 This simplified Herzsprung-Russell (H-R) diagram is based on data from thousands of stars. It shows that there is a relationship among the colour, temperature, luminosity, and mass of stars. **Inferring: Before deducing the patterns that led to the H-R diagram, most astronomers assumed that stars were unchanging and eternal. What can you see in the H-R diagram that challenges this assumption? What other assumptions could you make?**

main sequence a narrow band of stars on the H-R diagram that runs diagonally from the upper left (bright hot stars) to the lower right (dim cool stars)

The Main Sequence

The central band of stars stretching from the upper left to the lower right of the H-R diagram in Figure 4.32 is called the **main sequence**. The main sequence accounts for about 90% of the stars that you can see from Earth. What about the 10% of known stars *not* in the main sequence? Hertzsprung and Russell found that some stars were cooler but very luminous. The star Antares, for example, shown in Figure 4.33, has a surface temperature of only 3500°C, but it is the 15th brightest star in the night sky.

When placed on the H-R diagram, the cool, bright stars are far above the main sequence. Find these red giant and supergiant stars on the H-R diagram in Figure 4.32 and note their sizes. Table 4.3 summarizes some properties of main-sequence stars.

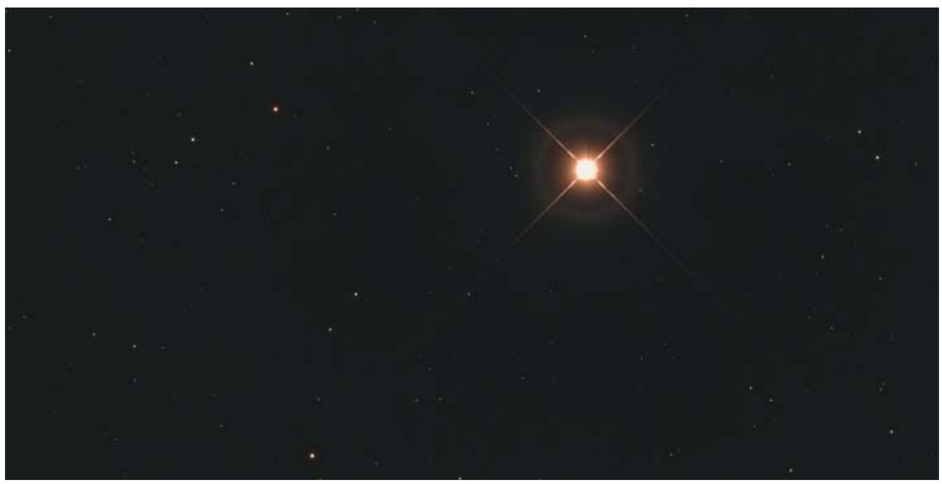


Figure 4.33 Antares is a bright supergiant located about 600 light-years from Earth. Its surface temperature is fairly cool, but it is extremely luminous.

Table 4.3 Some Properties of Main-Sequence Stars

Colour	Surface Temperature (°C)	Mass*	Luminosity*
blue	35 000	40	405 000
blue-white	21 000	15	13 000
white	10 000	3.5	80
yellow-white	7 500	1.7	6.4
yellow	6 000	1.1	1.4
orange	4 700	0.8	0.46
red	3 300	0.5	0.08

* relative to the Sun

Questions about why some stars are not in the main sequence led astronomers to wonder how these stars came to be. Were they special, rare types of stars that formed in a different way? Were they examples of main-sequence stars that had gone through dramatic changes at some stage in their life? Astronomers have worked out the basic features of stellar evolution. But, as you will read in the rest of this Concept, there are lots of details missing, and many puzzles still remain.

How Stars Evolve

Stars, in general, do not change very quickly. Stars like the Sun shine for billions of years with little or no change. Even so, stars radiate huge amounts of energy, and they cannot do that forever. Eventually, they run out of fuel. In the final stages of a star's life, it becomes a white dwarf, a neutron star, or a black hole. What the star evolves into depends on its initial mass on the main sequence. **Figure 4.34** outlines the evolution of different types of stars. Refer to this diagram as you read the text that follows.

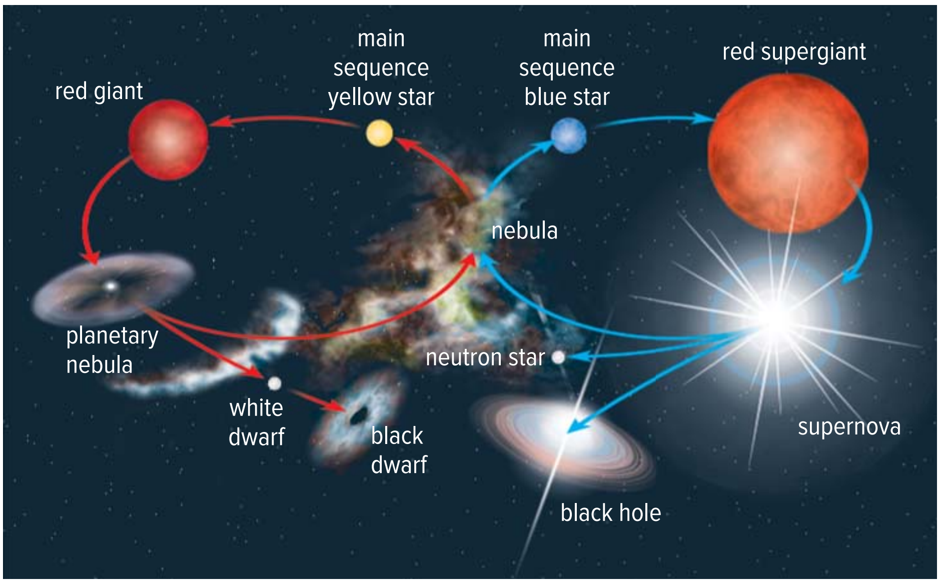


Figure 4.34 A star's life cycle depends on its initial mass.

Low-Mass Stars

Low-mass stars have less mass than the Sun. They consume their hydrogen slowly over a period that may be as long as 100 billion years. During that time they lose significant mass, essentially evaporating. In the end, all that remains of them is a very faint white dwarf. While white dwarfs no longer produce energy of their own, they are incredibly hot. It takes tens of billions of years for them to cool down. Astronomers theorize that when they do cool down, they will become nothing more than dark embers called black dwarfs. The universe is not old enough to contain any black dwarfs.

Intermediate-Mass Stars

Intermediate-mass stars, such as the Sun, consume their hydrogen over a period of about 10 billion years. When their hydrogen is used up, the core collapses, the temperature increases, and the outer layers begin to expand. The expanded layers are cooler and appear red. At this phase the star is called a red giant. In about five billion years the Sun will become a red giant. Its gaseous diameter will extend to the current orbit of Mars. Now called a planetary nebula, these gaseous layers will get more diffuse over time, and the Sun will eventually become a white dwarf.

High-Mass Stars

Stars that are 10 or more solar masses are high-mass stars. These stars consume their fuel even faster than the intermediate-mass stars. As a result, high-mass stars die more quickly and more violently. In massive stars, the core heats up to much higher temperatures. Heavier elements such as carbon and oxygen form by fusion, and the star expands into a supergiant. Eventually, iron forms in the core. Since iron cannot release energy through fusion, the core collapses violently, and a shock wave travels through the star. The outer portion of the star explodes, producing a **supernova**. A supernova can be millions of times brighter than the original star was. **Figure 4.35** shows the remains of a supernova.

supernova a massive explosion in which the whole outer portion of a star is blown off

During a supernova explosion, the heavier elements formed are ejected into the universe. Some of these elements become parts of new stars, and some form planets and other bodies. Your body and the life-sustaining environment surrounding you contain atoms that were fused in the cores of old stars.

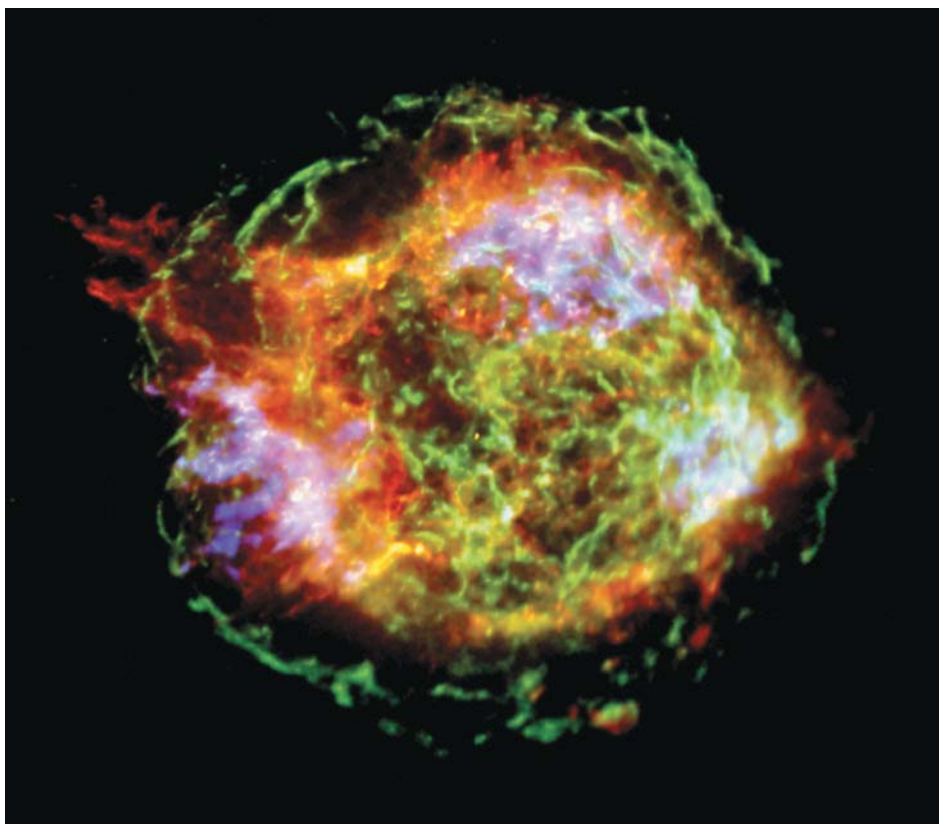


Figure 4.35 This is what is left of a star that exploded in the constellation Cassiopeia. Its light reached Earth 320 years ago after the star exploded about 11 000 light-years away in our Milky Way. **Analyzing: Elements that have been detected by the Chandra X-Ray Observatory, as well as earth-based equipment, include phosphorus, carbon, hydrogen, oxygen, and nitrogen. What is significant about these particular elements?**

Extending the Connections

Supernovae in Human History

Investigate supernovae that have been observed and recorded throughout human history. How reliable is the evidence that what people observed in each case was a supernova, rather than some other kind of celestial object or event?

Supernova Discovery

In 1987, Canadian astronomer Ian Shelton discovered a supernova while working at a University of Toronto observatory in Chile. Shelton was examining images of stars, and he noticed something unusual in one of them. He decided to step outside and look up. Among thousands of stars he spotted a bright one that was not previously visible. See **Figure 4.36**. Shelton had discovered a supernova. Called SN 1987A, the supernova was the closest one to Earth since 1604. SN 1987A is 163 000 light-years from Earth and is the only one visible to the unaided eye.

Depending on the initial mass of the star before it became a supernova, the remaining star will become either a neutron star or a black hole.

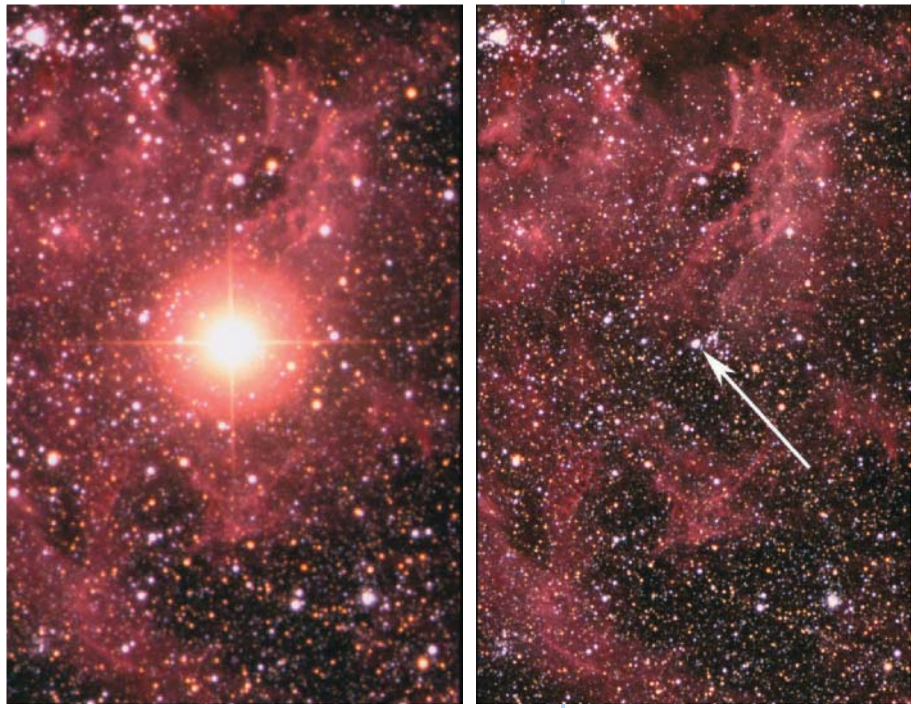


Figure 4.36 The photo on the left shows the supernova discovered by Ian Shelton. The photo on the right shows the same area before the recorded supernova.

Activity

Modelling a Supernova

When the core of a star collapses, it does so with such enormous force that it rebounds. As the core collapses, all the outer atmospheric layers are also collapsing. The less dense outer layers are still falling in toward the core when the core rebounds. The rebounding core collides with the outer layers with enough energy to blow the atmospheric layers away from the star. This is the supernova explosion. In this outdoor activity, a basketball models the core, and a tennis ball models the star's outer atmospheric layers.

1. Drop the basketball and then the tennis ball. Record your observations.
2. Place the tennis ball on top of the basketball, and hold them out in front of you. Predict how each ball will bounce if you release them at the same time. Then test your prediction.
3. How do your observations in step 1 compare with step 2?
4. What is the source of the extra energy that caused the result you observed?
5. How is this model like a supernova explosion?
6. What parts of this model are not like a supernova explosion?

Figure 4.37 The Crab Nebula is the remains of a supernova observed in 1054. Chinese historical records from that year referred to it as a "guest star."

Neutron Stars

If the star began with a mass of about 10 solar masses, its core will shrink to about 20 km in diameter. In such stars, the pressure is so great that electrons combine with protons to become neutrons, and the star eventually becomes a **neutron star**. The first neutron star to be discovered is in the centre of the Crab Nebula (**Figure 4.37**). Astronomers have discovered that it is spinning about 30 times per second. As it spins, it sends out pulses of radiation. This neutron star was among the first discoveries of a type of neutron star called a pulsar, which send out pulses of radiation, much like an extremely fast-sweeping searchlight.

neutron star a star so dense that only neutrons can exist throughout

black hole the remnant of a supernova explostion with a gravitational field so strong that nothing can escape its pull

Black Holes

The most spectacular deaths happen to stars whose initial masses are more than 20 solar masses. The remnant of the supernova explosion is so massive that nothing can compete with the crushing force of gravity. The remnant is crushed into a black hole. A **black hole** is a tiny patch of space that has no volume, but it does have mass. Therefore, there is still gravity. In fact, the gravitational force of a black hole is so strong that nothing can escape it. Even light cannot escape a black hole's gravity.

Black holes are among the strangest objects in the universe. Astronomers predicted the existence of black holes before the first one was discovered. Recall how Mendeleev's periodic table modelled the elements that were known at the time. Mendeleev's model led to predictions of missing elements, which then led to discoveries. In a similar way, scientists build mathematical models of how stars evolve and eventually die. The models seemed to fit what scientists were seeing, so when the models pointed to the possibility that a strange object like a black hole could exist, scientists started looking for them.

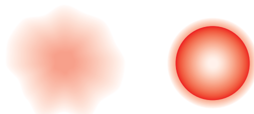
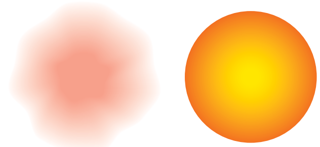
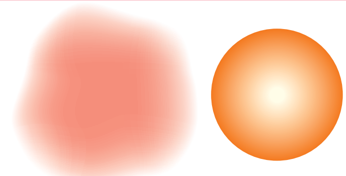
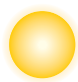
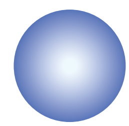
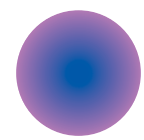
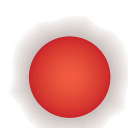
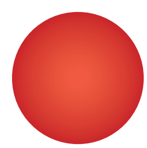
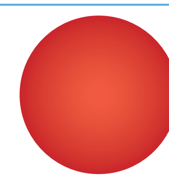
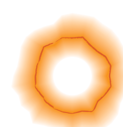
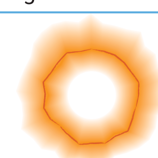
When astronomers say they have detected a black hole, they do not mean that they have seen one. What they mean is that they have detected the gravitational effects of an object whose mass and size match those predicted by physics. For example, black holes that exist in congested regions of space sometimes swallow up matter and compress it until enormous temperatures are reached before it disappears. When this happens, the black hole emits intense radiation that uncloaks the normally invisible black hole. Astronomers look for the telltale signature of black holes devouring gas, dust, and stars using radio, X-ray, and gamma ray telescopes.

In 1972, Dr. Tom Bolton, of the University of Toronto, identified the first black hole. It is called Cygnus X-1 and is located in our Milky Way. Astronomers now know that, in the centre of most galaxies there are supermassive black holes with a mass millions of times that of the Sun's.

Summarizing the Life Cycles of Stars

Figure 4.38 summarizes the life cycle of stars, based on mass.

Figure 4.38 Life cycle pathways for stars based on their mass. (The drawings are not to scale.)

Star	Low-Mass (<5 solar masses)	Intermediate-Mass (10–20 solar masses)	High-Mass (>20 solar masses)
Birth and early life	 Forms from a small- or medium-sized portion of a nebula; gradually turns into a hot, dense clump that begins producing energy.	 Forms from a large portion of a nebula, and in a fairly short time turns into a hot, dense clump that produces large amounts of energy.	 Forms from an extremely large portion of a nebula, very quickly turning into a hot, dense clump that produces very large amounts of energy.
Main sequence phase	 Uses nuclear fusion to produce energy for about 10 billion years if the mass is the same as the Sun's, or 100 billion years or more if the mass is less than the Sun's.	 Uses nuclear fusion to produce energy for only a few million years. It is thousands of times brighter than the Sun.	 Uses nuclear fusion to produce energy for only a few million years. It is extremely bright.
Old age	 Uses up fuels and swells to become a large, cool red giant.	 Uses up fuels and swells to become a red supergiant.	 Uses up fuels and swells to become a red supergiant.
Death	 Outer layers of gas drift away, and the core shrinks to become a small, hot, dense white dwarf star.	 Core collapses; outer layers explode as supernova.	 Core collapses; outer layers explode as huge supernova.
Remains	 White dwarf star eventually cools and fades.	 Core material packs together as a neutron star. Gases drift off as a nebula to be recycled.	 Core material packs together as a black hole. Gases drift off as a nebula to be recycled.

Before you leave this page . . .

1. List and briefly describe four properties of stars.
2. Which two properties of stars are compared in an H-R diagram?
3. What does an H-R diagram help us understand about stars?
4. Use a sketch or flowchart to outline how a star's life cycle depends on its initial mass.

INVESTIGATION

4-C

STRUCTURED INQUIRY

Skills and Strategies

- Questioning and Predicting
- Processing and Analyzing
- Evaluating
- Communicating

Build an H-R Diagram

In this investigation, you will build an H-R diagram from the data in the table and use it to predict the absolute magnitudes of main-sequence stars.

Question

How can you predict the absolute magnitude of stars in the main sequence if you know its spectral type?

Procedure

1. Your teacher will give you a blank H-R diagram. To make it easier to refer to main-sequence stars, astronomers developed a series of star types based on their spectra. These spectral types are named O, B, A, F, G, K, and M. These are further broken down into subgroupings with numbers, such as O5 and B2. Using the table of data, graph each spectral class against its absolute magnitude.

Spectral Type	O5	O9	B0	B2	B5	A0	A2	F0	F2
Absolute Magnitude	−5.7	−4.5	−4.0	−2.45	−1.2	+0.65	+1.3	+2.7	+3.6

Spectral Type	F8	G2	G8	K0	K2	K5	M0	M2	M5
Absolute Magnitude	+0.44	+4.7	+5.5	+5.9	+6.4	+7.35	+8.8	+9.9	+12.3

Analyze and Interpret

1. Describe the pattern on your graph.
2. The Sun's spectral type is about G2. Use your chart to predict the Sun's absolute magnitude.
3. Vega's spectral type is A0. Predict Vega's absolute magnitude.
4. Predict the absolute magnitude of a star whose spectral type is B5.

Conclude and Communicate

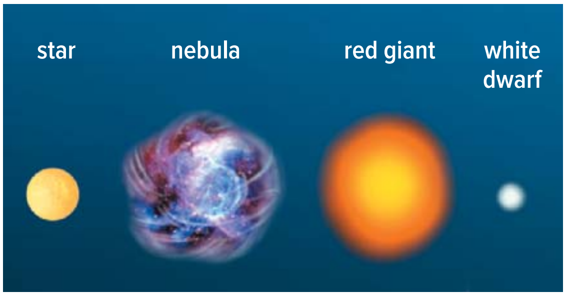
5. Summarize how you can use this pattern to predict a main-sequence star's absolute magnitude if you know its spectral type.

Check Your Understanding of Topic 4.3

QP Questioning and Predicting PC Planning and Conducting PA Processing and Analyzing E Evaluating AI Applying and Innovating C Communicating

Understanding Key Ideas

1. How can the temperature of a star be inferred from its physical properties? PA
2. Which types of stars are thought to be the sources for enriching the universe with heavy elements? Explain why. PA
3. Place the following from youngest to oldest. Explain your sequence. PA



4. Can a star such as our Sun become a supernova? Explain why or why not. PA E
5. How can the evolution of a star over its life cycle be tracked using an H-R diagram? Use a specific example of a category of star. PA C
6. How far does light travel in one second? in one nanosecond? PA A
7. What are black holes, and why are they considered to be unusual? PA C

Connecting Ideas

8. Stars are grouped into seven colour types, and each is identified by a letter: O, B, A, F, G, K, or M. Use the table below, as well as **Figure 4.38**, and what you have learned about star evolution to answer the following questions. Explain your answer in each case. E AI C
   a) Which star type is a red dwarf?
   b) Which star type is most like the Sun?
   c) Which star types are likely to become supernovae? Why? How long will it be before this occurs?
   d) Our Sun is halfway through its life on the main sequence. If all seven stars formed at the same time as the Sun, which ones could we observe today?

Star Type	Colour	Lifetime on Main Sequence (Years)
O	blue	1 million
B	blue-white	11 million
A	white	440 million
F	yellow-white	3 billion
G	yellow	8 billion
K	orange	17 billion
M	red	56 billion

Making New Connections

9. Galaxies can collide with each other. This photo shows the Antennae galaxies as they are colliding. E AI C QP

   a) Galaxy collisions are common, but it is unlikely that stars within two colliding galaxies will strike each other. Infer why.
   b) Galaxy collisions can spawn the birth of stars. Infer why.
   c) Our galaxy is predicted to collide with the Andromeda Galaxy in the next 4 billion years. Predict how Earth would be affected by this.

AT ISSUE How did Jocelyn Bell make one of the most important astronomy discoveries of the 20th century?

What's the Issue?

Before neutron stars were discovered, the smallest and most dense stars were thought to be white dwarf stars. In fact, until the 1960s, most astronomers thought that all stars in the late stages of their life cycles would eventually become white dwarf stars. However, 30 years earlier, two astronomers had cautiously proposed the idea of an even smaller, denser object. "With all reserve," they began, "we advance the view that a super-nova represents the transition of an ordinary star into a new form of star, the *neutron star*, consisting mainly of neutrons. Such a star may possess a very small radius and an extremely high density." (This scientific paper, published in 1934 by Walter Baade and Fritz Zwicky, included the first published use of the term "supernova.")

With the gauntlet dropped, some astronomers started to search for neutron stars. One region of space that was studied closely was the Crab Nebula, because it was the result of a known supernova in 1054. If neutron stars existed, the centre of the Crab Nebula should yield a discovery. It did not, however, and astronomers mostly lost interest in seeking the hypothesized—and quite possibly fictional—neutron star.

The stellar landscape changed in 1967, due to a chance discovery by 24-year-old PhD student named Jocelyn Bell. There is a saying that "chance favours the prepared mind," and it is in the spirit of this saying that the word "chance" was used in the previous sentence. As part of her work as a research assistant, Bell had helped to build a large radio telescope at the Mullard Radio Astronomy Observatory in Cambridge, England. The aim of the work was to study quasars, and Bell was tasked with analyzing the reams of data collected by the telescope. (The data was in the form of huge rolls of chart paper—"red pen over moving paper", as Bell put it in her Presidential Address of 2003—

Artist's conception of a neutron star

which would then be analyzed. “In the six months that I personally operated the telescope,” she said, “several miles of chart were recorded.”

Bell had noticed a tiny anomaly—a quarter of an inch of what she called a “scruffy signal” in every 400 feet of the chart paper. The interval between each of these radio signals was a constant 1.3 seconds. No celestial object was known to emit a signal like this. Bell and her research associates were stumped. They took to calling these signals “little green men” and for a time used the abbreviation LGM to identify the signals on the chart paper.

By late December of 1967, Bell had found three more pulsing radio sources, which they gave the name *pulsars* in an interview with a newspaper journalist. The following year, a pulsar was discovered at the centre of the Crab Nebula. Based on this and other evidence linking pulsars with supernovas, Bell and other astronomers concluded that pulsars and the long-hypothesized neutron stars were the same thing. Since then, about 2000 pulsars have been identified.

Jocelyn Bell at the Mullard Radio Astronomy Observatory in 1968. She’s holding a roll of the chart paper used to record the signals that led to the discovery of pulsars.

Dig Deeper

Collaborate with your classmates to explore one or more of these questions—or generate your own questions to explore.

1. The regularity of the pulsating radio signals led Bell and her associates to infer, early on, that the signals must be coming from a human-made source. What reasoning would lead to such an inference?
2. Jocelyn Bell’s advisor, Anthony Hewish, and another colleague of his were awarded the Nobel Prize for Physics in 1974 for the discovery of pulsars. Bell was excluded from the reward, despite her principal role in the discovery and her name on the scientific paper that first presented it. How commonplace was it for the role of women in scientific achievement to be ignored or downplayed? How has the situation changed? Is there still more change necessary?
3. Find interviews and addresses that Jocelyn Bell has made since 1967. How does the process of scientific inquiry that you have been learning about in school compare with how she describes and characterizes it? (Note: Most references to Jocelyn Bell that you will encounter use her married name: Jocelyn Bell Burnell.)